5. Using the following data, construct Fisher's Ideal Index Number and Show that it satisfies Factor Reversal Test and Time Reversal Test?

Commodities	Price		Quantity	
	Base Year	Current Year	Base Year	Current Year
Wheat	6	10	50	56
Ghee	2	2	100	120
Firewood	4	6	60	60
Sugar	10	12	30	24
Cloth	8	12	40	36

6. Compute the consumer price index for 2015 on the basis of 2014 from the following data.

Commodities	Quantities	Prices in 2015	Prices in 2016
A	6	5.75	6.00
B	6	5.00	8.00
C	1	6.00	9.00
D	6	8.00	10.00
E	4	2.00	1.50
F	1	20.00	15.00

7. An Enquiry was made into the budgets of the middle class families in a city gave the following information.

Expenditure	Food	Rent	Clothing	Fuel	Rice
Price(2010)	150	50	100	20	60
Price(2011)	174	60	125	25	90
Weights	35	15	20	10	20

What changes in the cost of living have taken place in the middle class families of a city?

8. From the following data, calculate the control limits for the mean and range chart.

Sample No.	1	2	3	4	5	6	7	8	9	10
Sample Observations	50	51	50	48	46	55	45	50	47	56
	55	50	53	53	50	51	48	56	53	53
	52	53	48	50	44	56	53	54	49	55
	49	50	52	51	48	47	48	53	52	54
	54	46	47	53	47	51	51	57	54	52

XII Std - Business Maths & Stat EM Chapter 9.indd 232 2/28/2020 5:11:36 PM